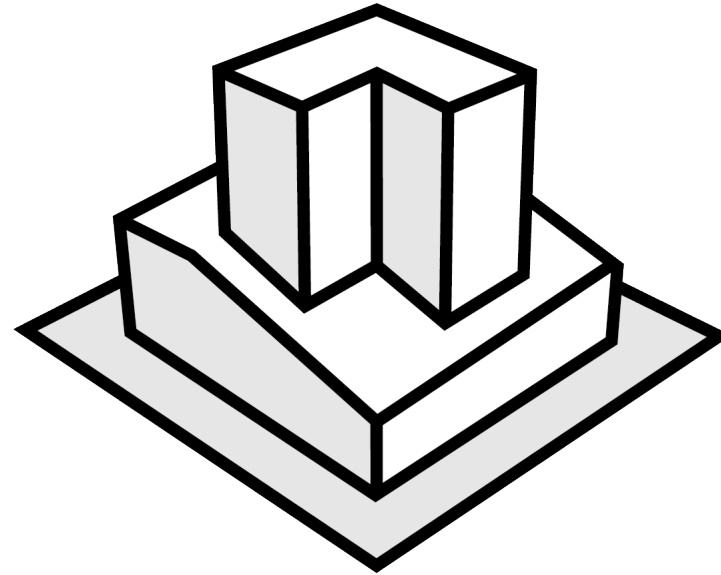
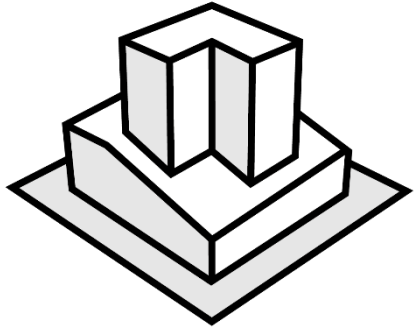


SDU Center for Large Structure Production (LSP)

Robotization and digitalization
for the sectors maritime, construction & energy

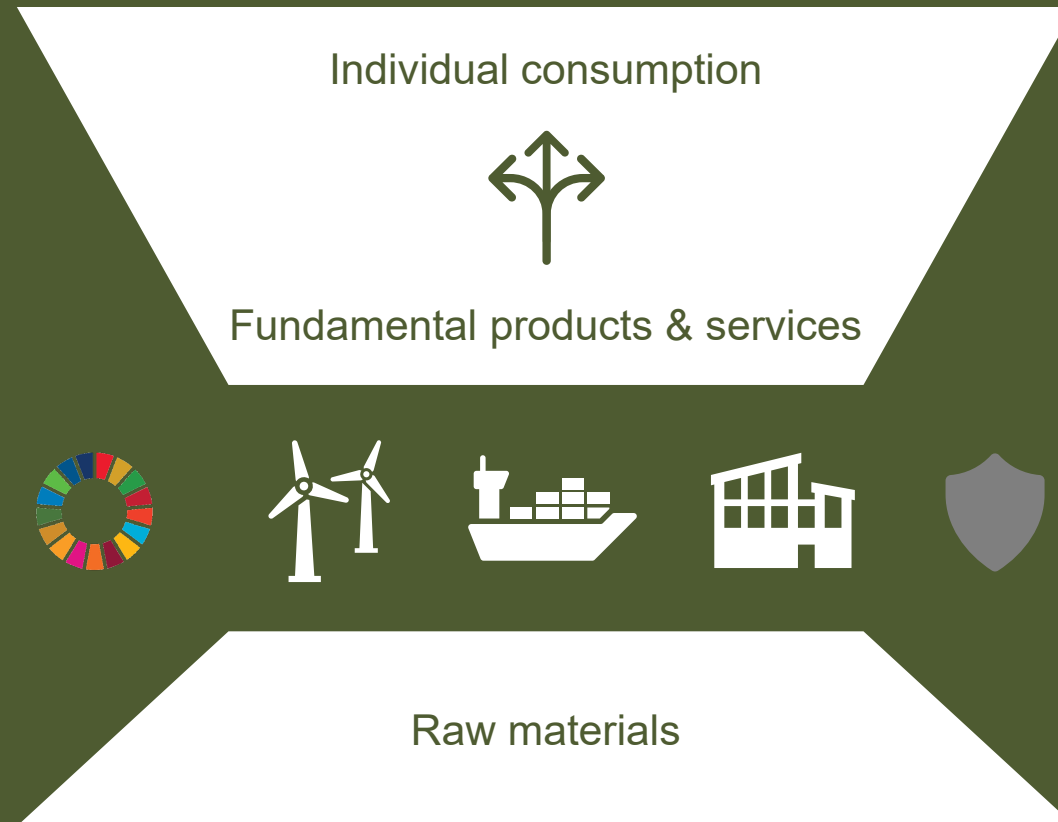




LSP Quick intro



LSP addresses today's production challenges





Fayard



CS Wind
Offshore



Status today

Manual labor — Production of large structures is still mainly carried out manually today

Global competition — Due to labor costs, production competencies are leaving Denmark

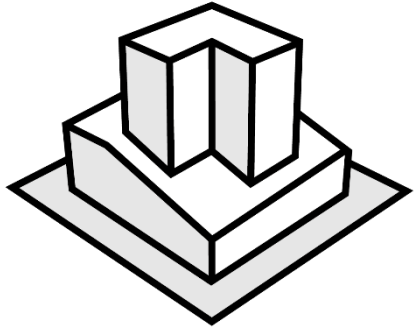
Lack of workforce — What remains as production is lacking young talents

Lack of automation — Robotics and AI could compensate but are not available

Lack of research — Due to equipment costs, robotics research is not dedicated to LSP

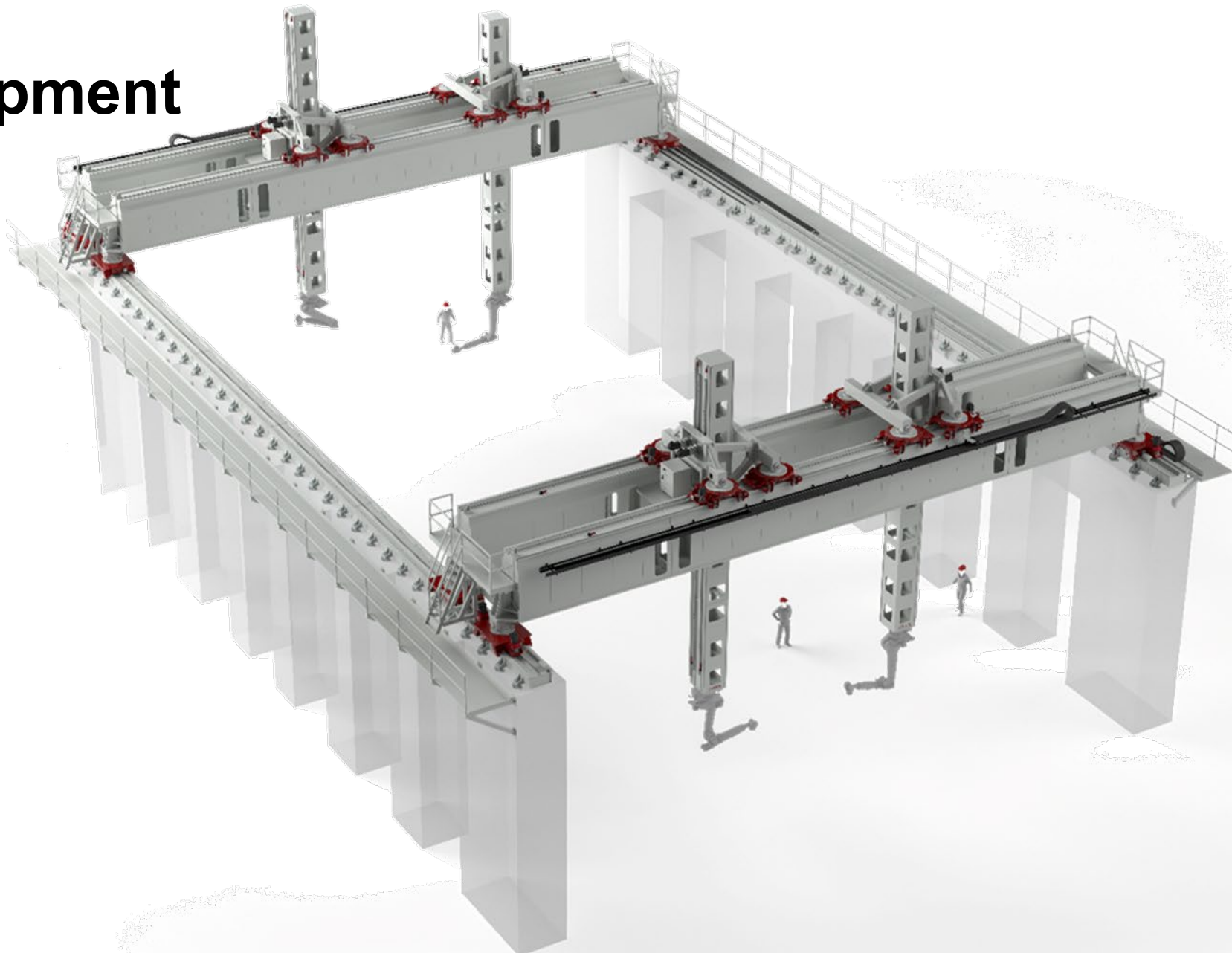
LSP partners

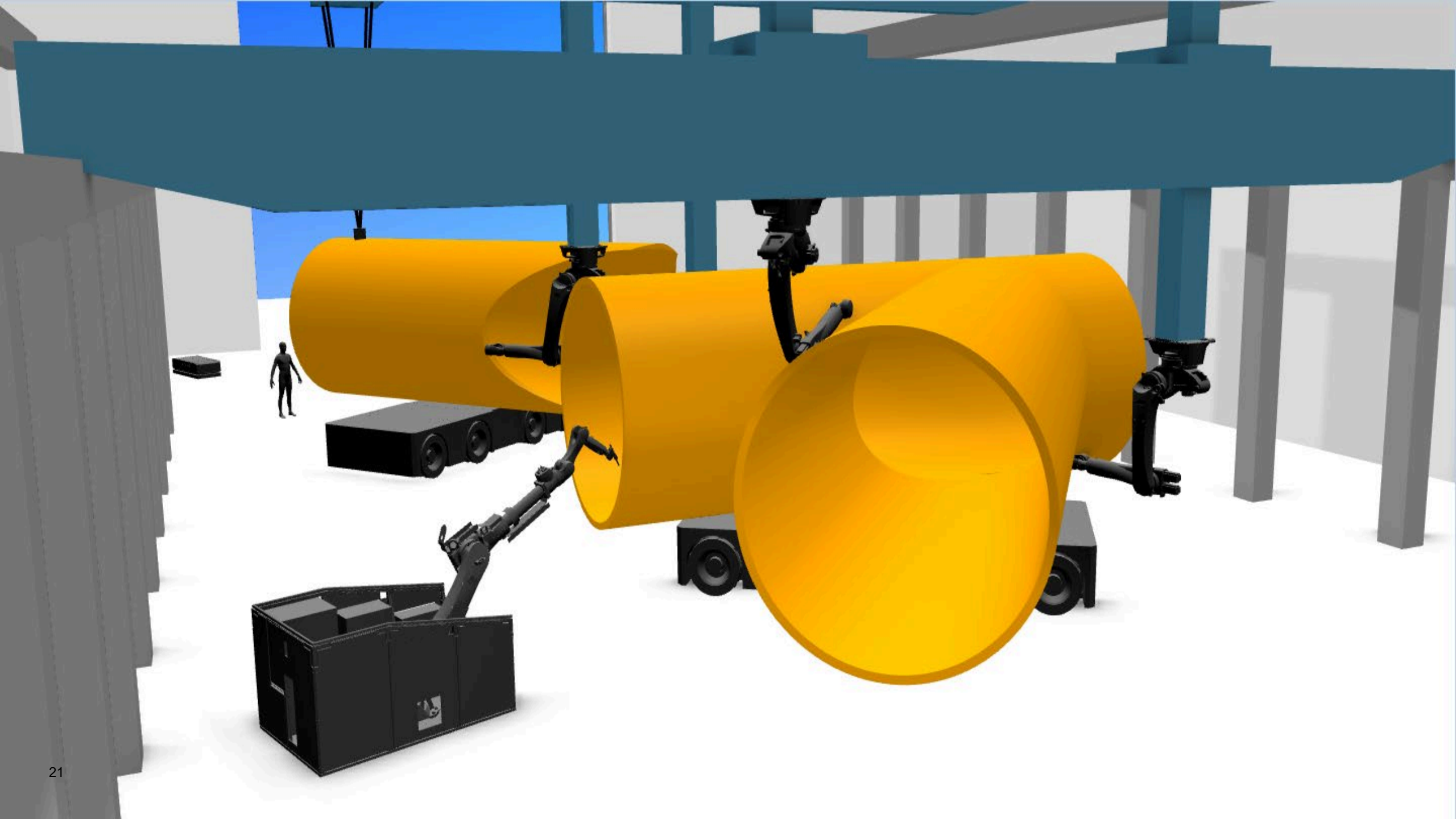
Partners	Collaborators in Use Cases	Interest from the sectors	Stakeholders / automation community	Collaborators / cluster community	Partners / education community
   	                    	               	             	       	   

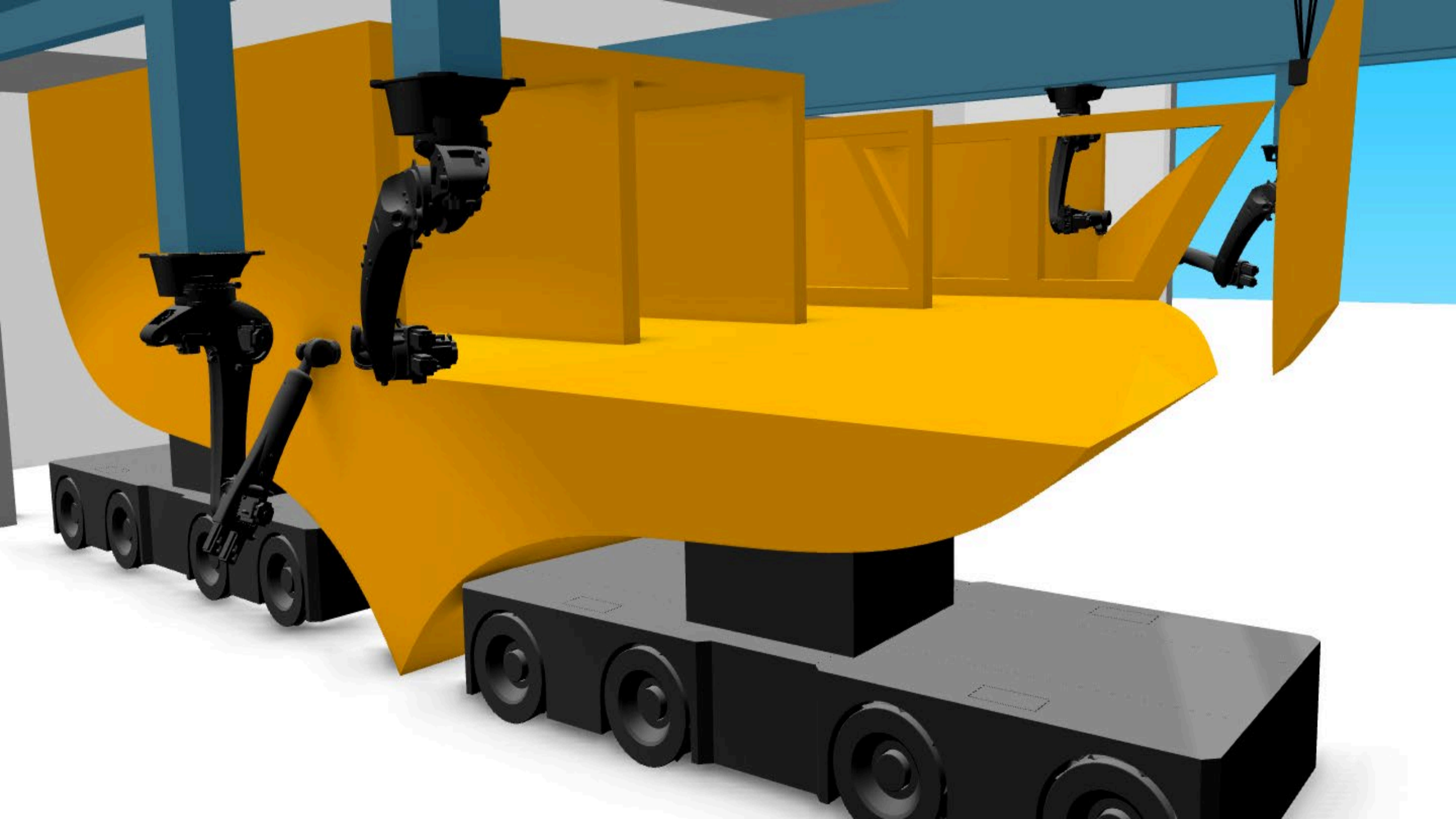


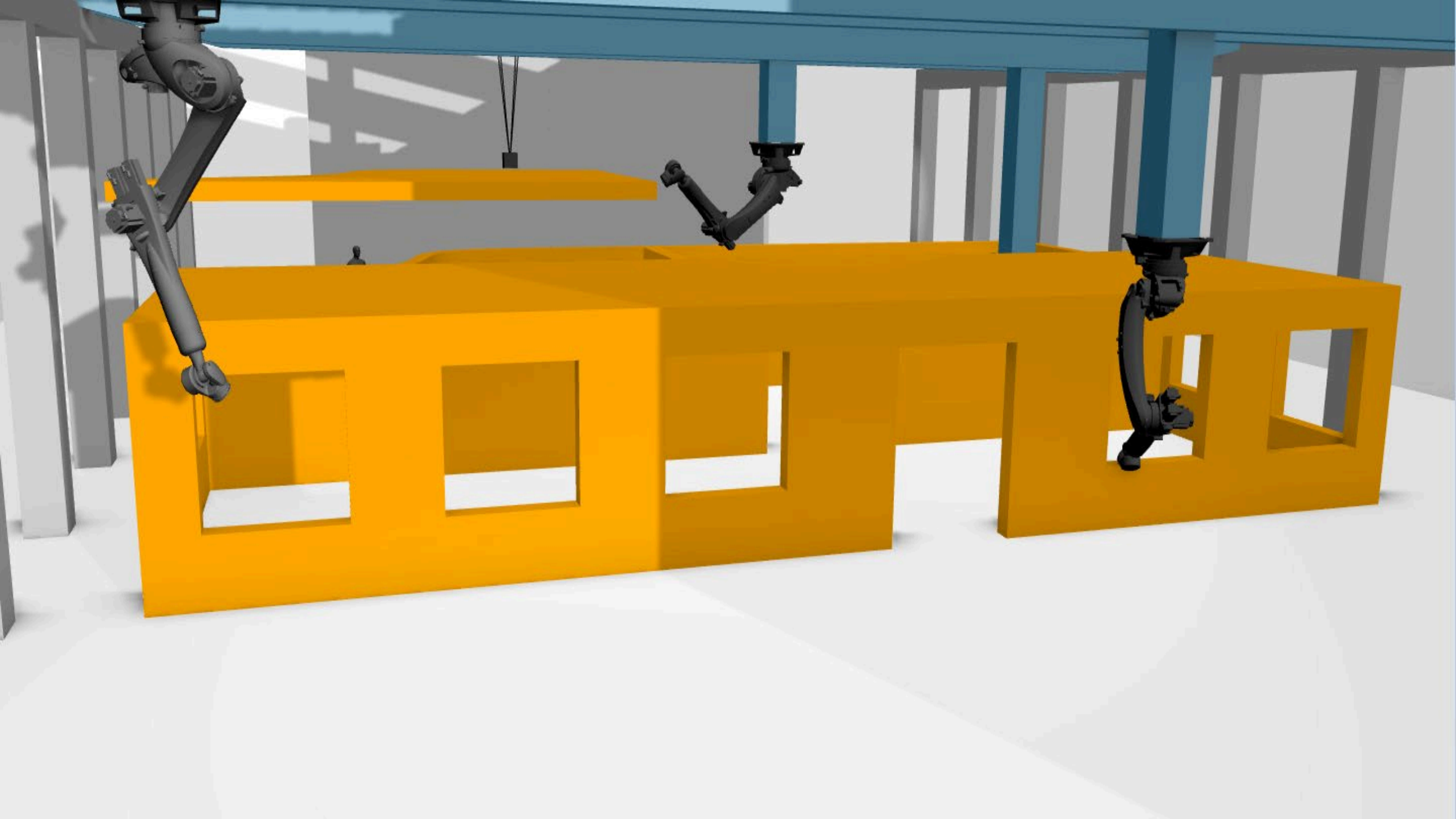
LSP Equipment

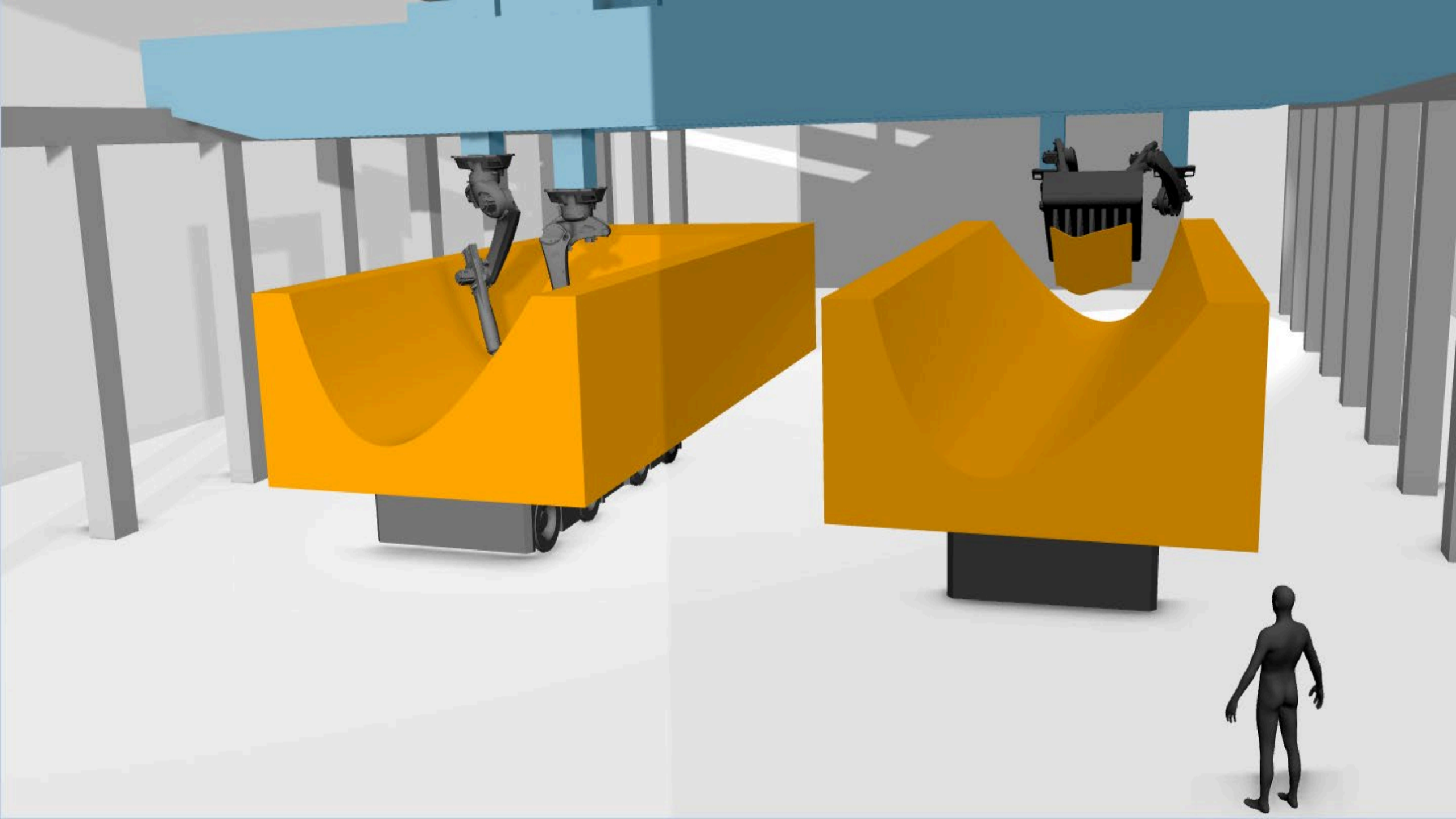
LSP equipment

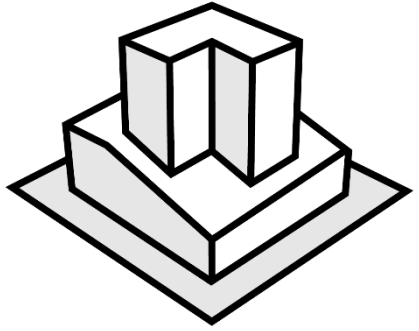












LSP Network

#largestructureproduction

What's in for companies?

Develop new systems and services with LSP

Become partner in an existing use case

Define your own project and consortium

Engage in project for funding

Order LSP services to use LSP equipment

Engage in the LSP platform

Meet the sectors construction, maritime and energy

Meet the Odense Robotics community and the automation sector

Learn about technologies and upskill your staff

Present your products and services

LSP Services

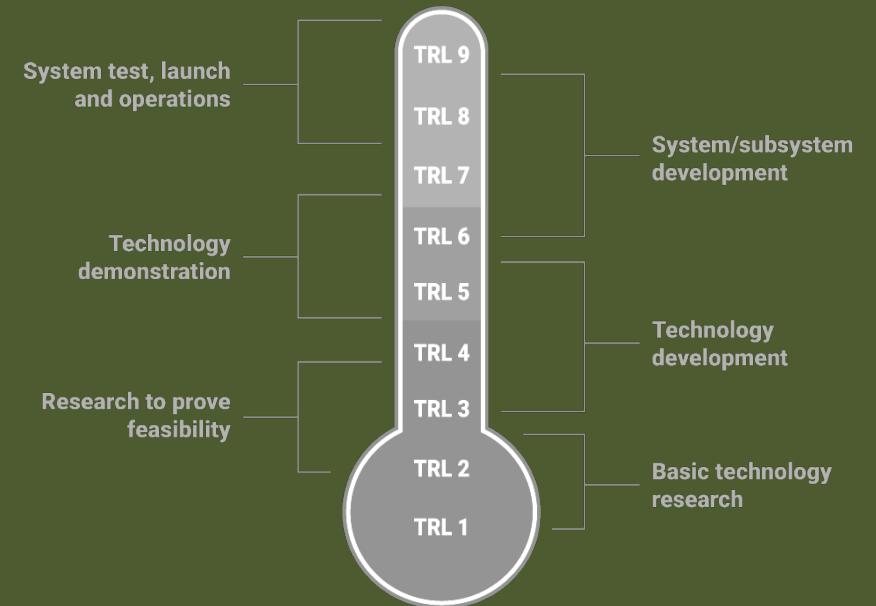
Feasibility & consultancy – TRL 1-9

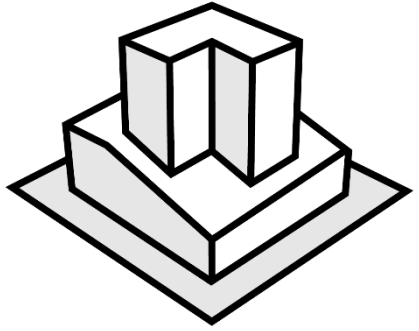
Research & development – TRL 1-9

Testing & experimentation – TRL 5-8

Low volume production – TRL 8+9

Education & training – TRL 8+9

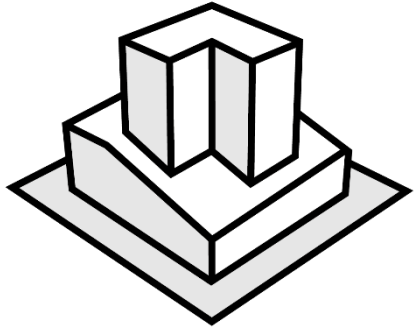




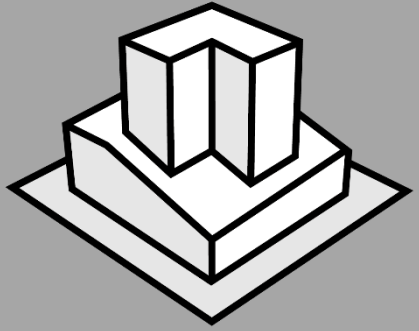
LSP Building





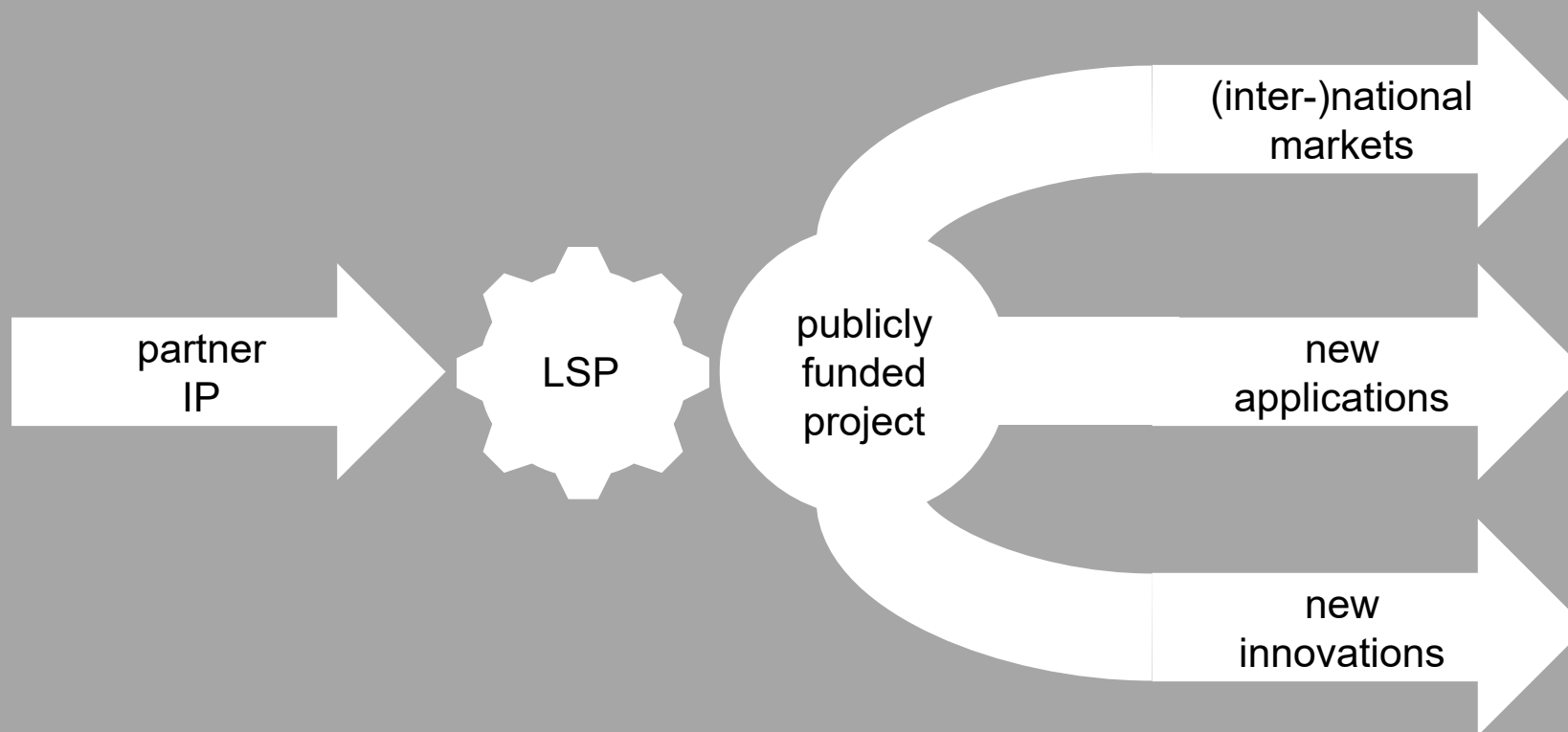
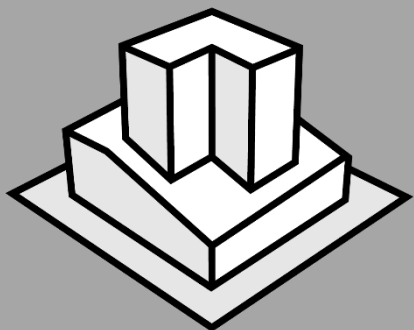


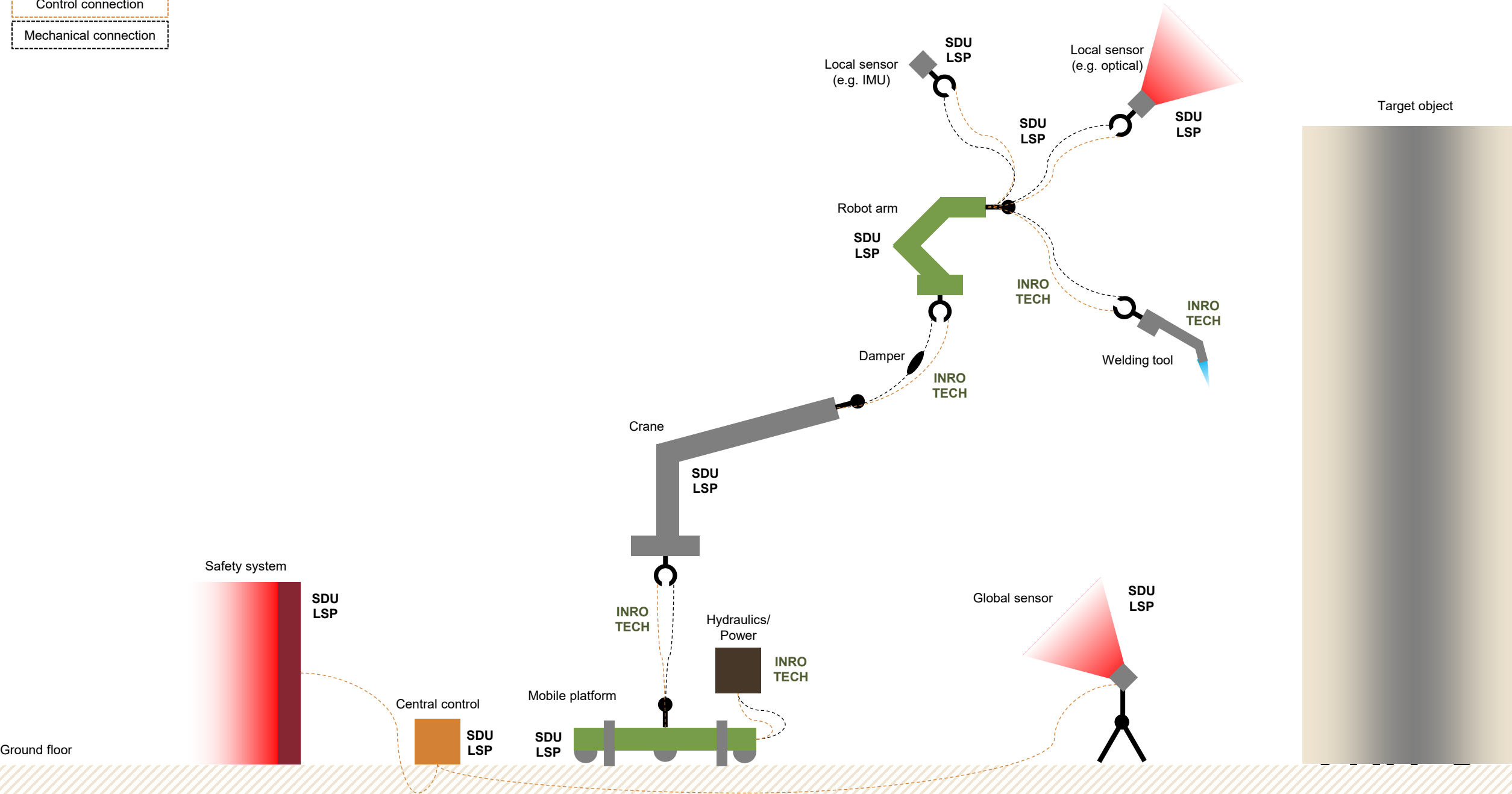
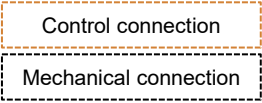
LSP Projects



Mobile Robot Crane Platform

EUDP

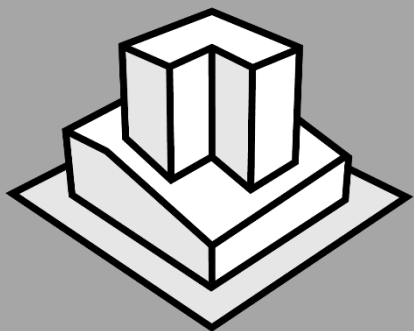






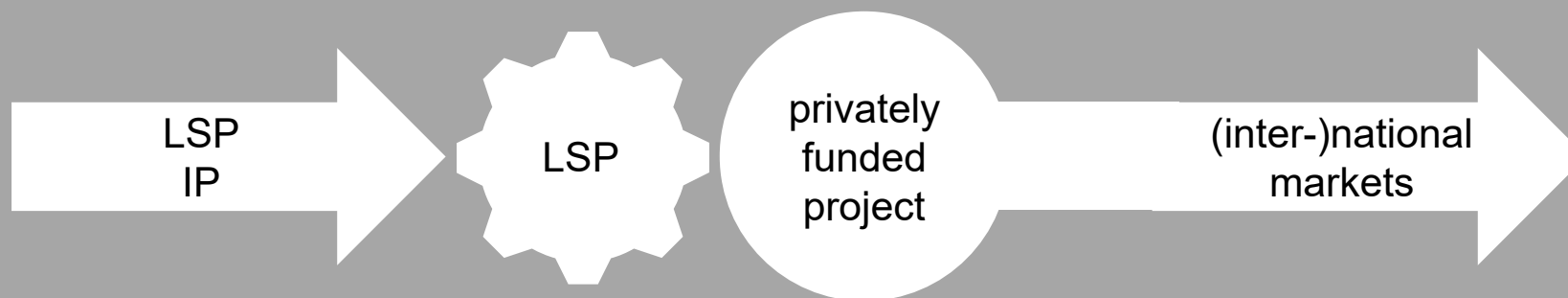
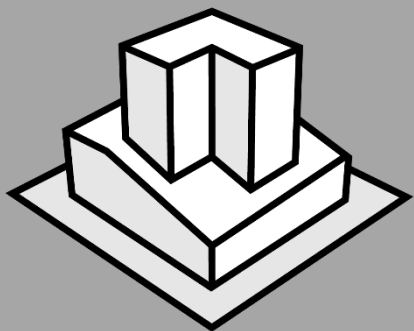






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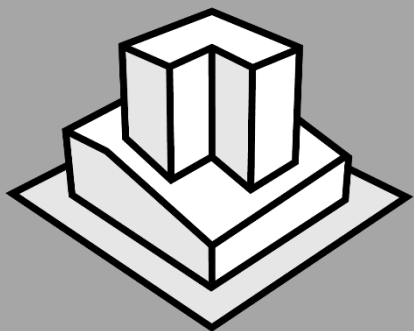
PERI











LSP Ship Factory

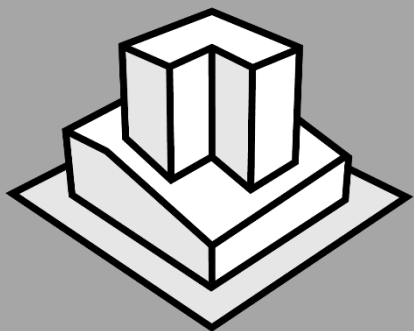
IFD

**The joint development of novel technologies
for the highly automated building, repair and recycling
of high value, modular ships in Denmark.**



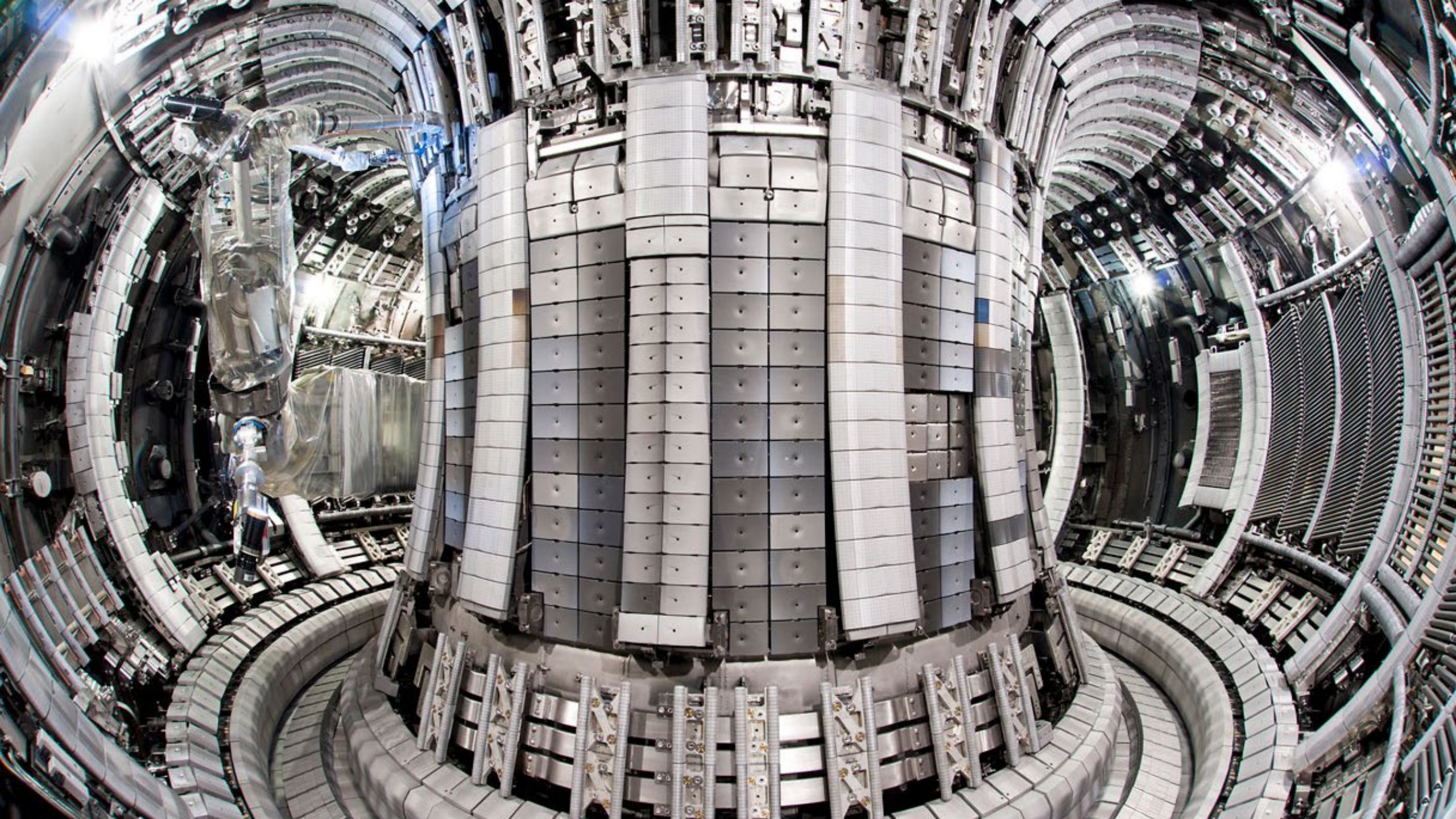
Consortium

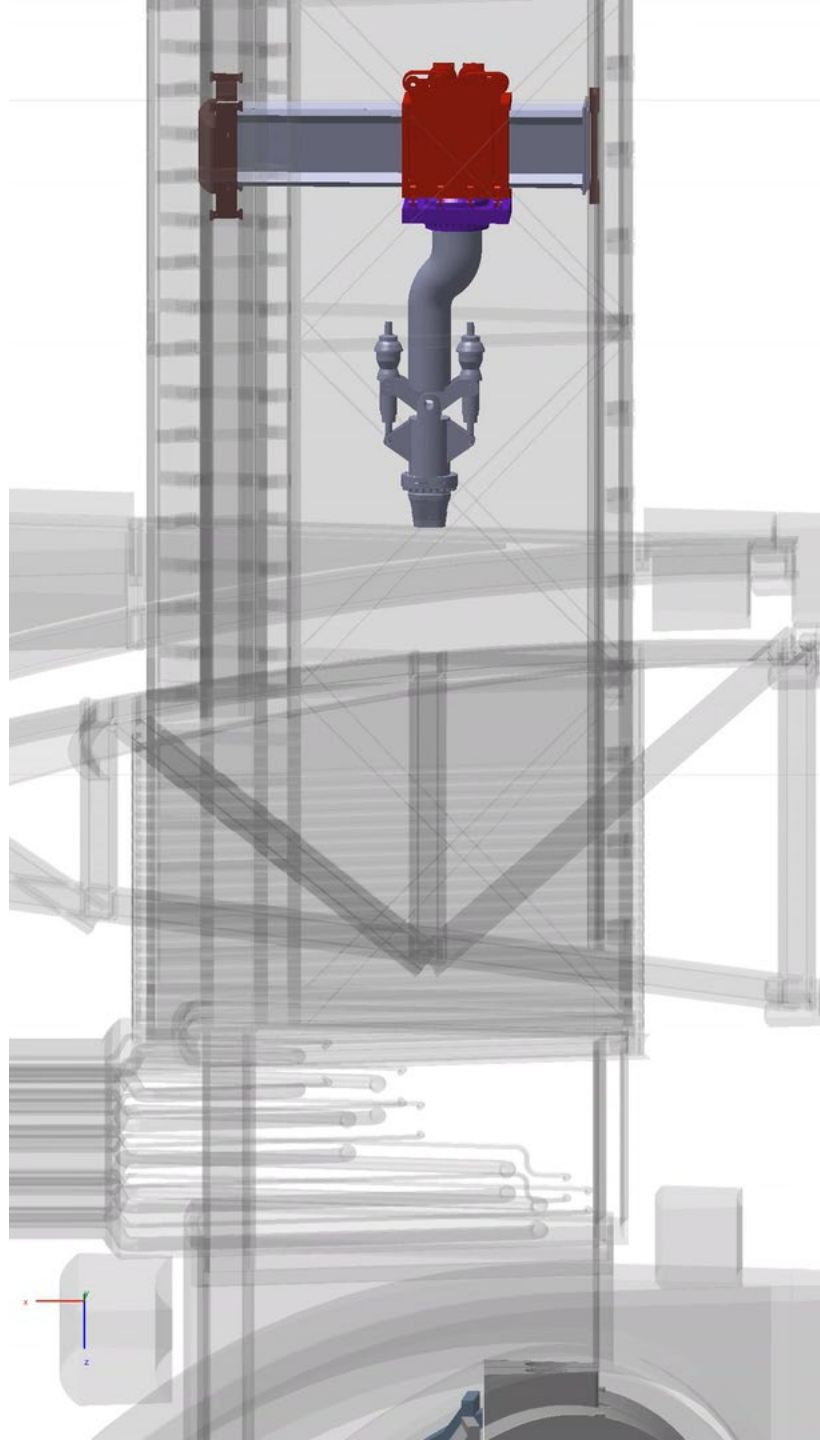
Partner		Type	Expertise
	SDU Center for Large Structure Production (LSP)	University	Coordination / Ship Factory control architecture development and integration
	Danske Maritime	Non-profit	Industrial coordination / Dissemination in maritime sector
	Inrotech A/S	Company	Robotic welding supplier
	FORCE Technology	RTO	Welding and additive steel manufacturing
	Simfact ApS	SME	Industrial robot simulation
	Fayard A/S	Company	Shipyard
	Orskov Yard A/S	Company	Shipyard
	Karstensens Skibsværft A/S	Company	Shipyard
	Søby Shipyard A/S	Company	Shipyard
	Tuco Marine Group A/S	Company	Shipyard
	SDU Mechanical Engineering	University	Ship design optimization
	Odense Maritime Technology A/S	Company	Ship design and shipyard consultancy
	Hauschildt Marine A/S	Company	Ship design
	SH Group A/S	Company	Modular maritime equipment
	Odense Robotics	Cluster	Dissemination in automation sector



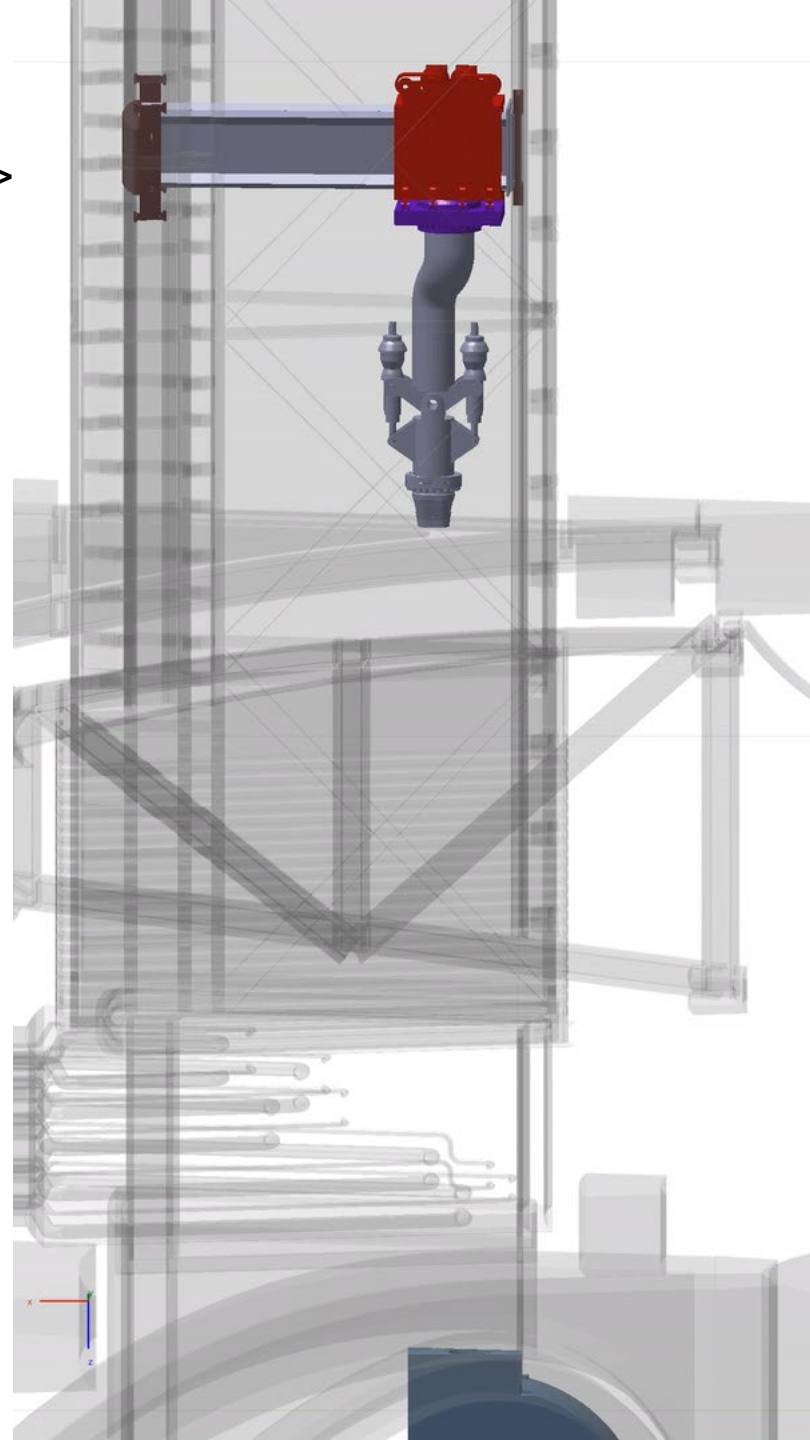
Fusion BB RH Facility

EUROfusion





< Outer blanket
Inner blanket >



BB Remote Handling Facility

A globally unique facility embedded in international R&D

Development of robotic capabilities relevant for fusion and large structure production

Challenge:

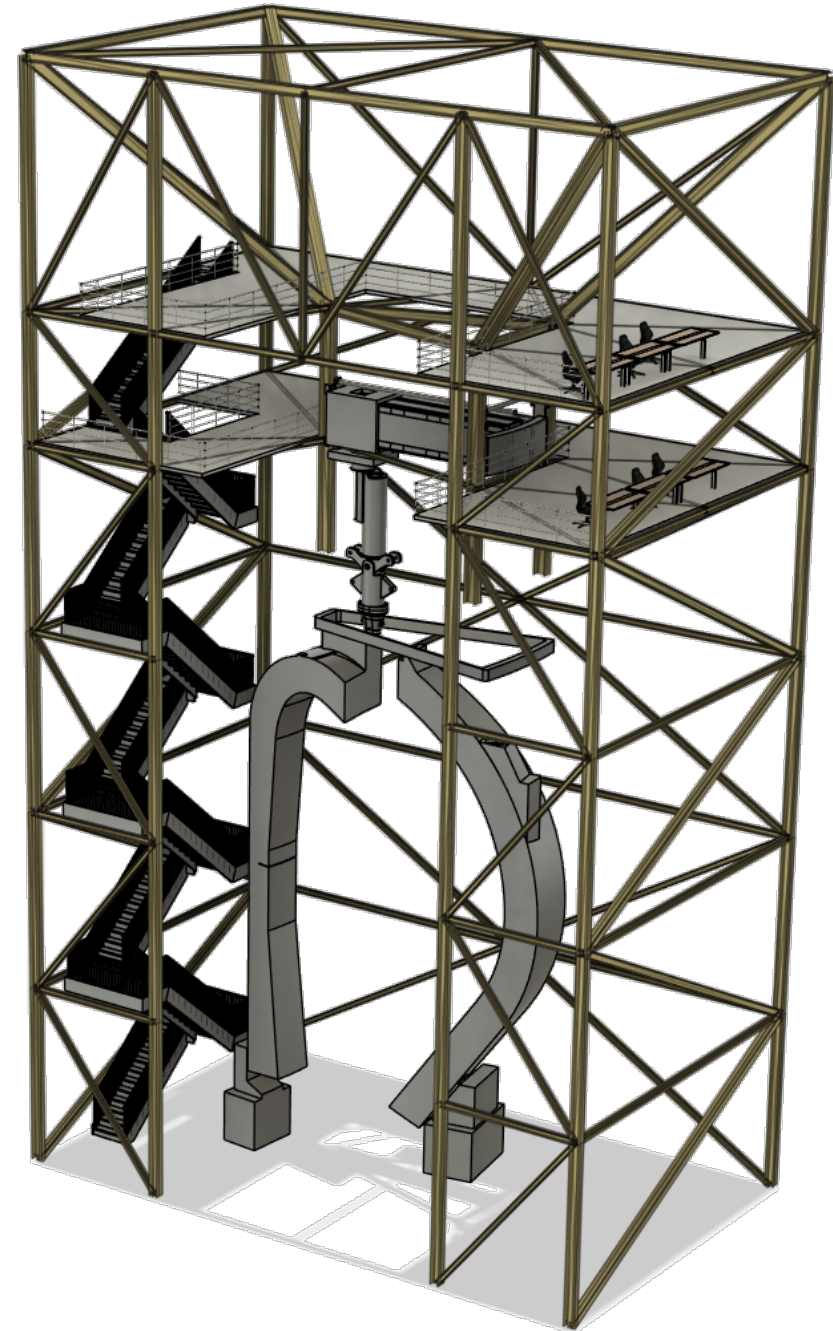
- 90 t payload handling with single robotic manipulator
- Payload flexes under own weight and subject to varying forces
- Small clearances with no collisions tolerated

Solution:

- Remote robotic manipulation
- driven by dynamic digital twin and sensors

Separate facility at LSP (funded by EUROfusion)

- Dimensions B x W x H = 10 m x 8 m x 25 m
- Detailed design by SDU in 2024/25
- Construction after EU tender 2025/26
- Commissioning in 2026
- Industrial capacity building from 2027 onwards





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Professor

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